ASSIGNMENT ONE

Python Basics

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`Main file: assignment 1_yourname.py`

GitHub repository

https://github.com/21echo12/assignment1-python-basics

01 Turtle Drawing - Prism Garden

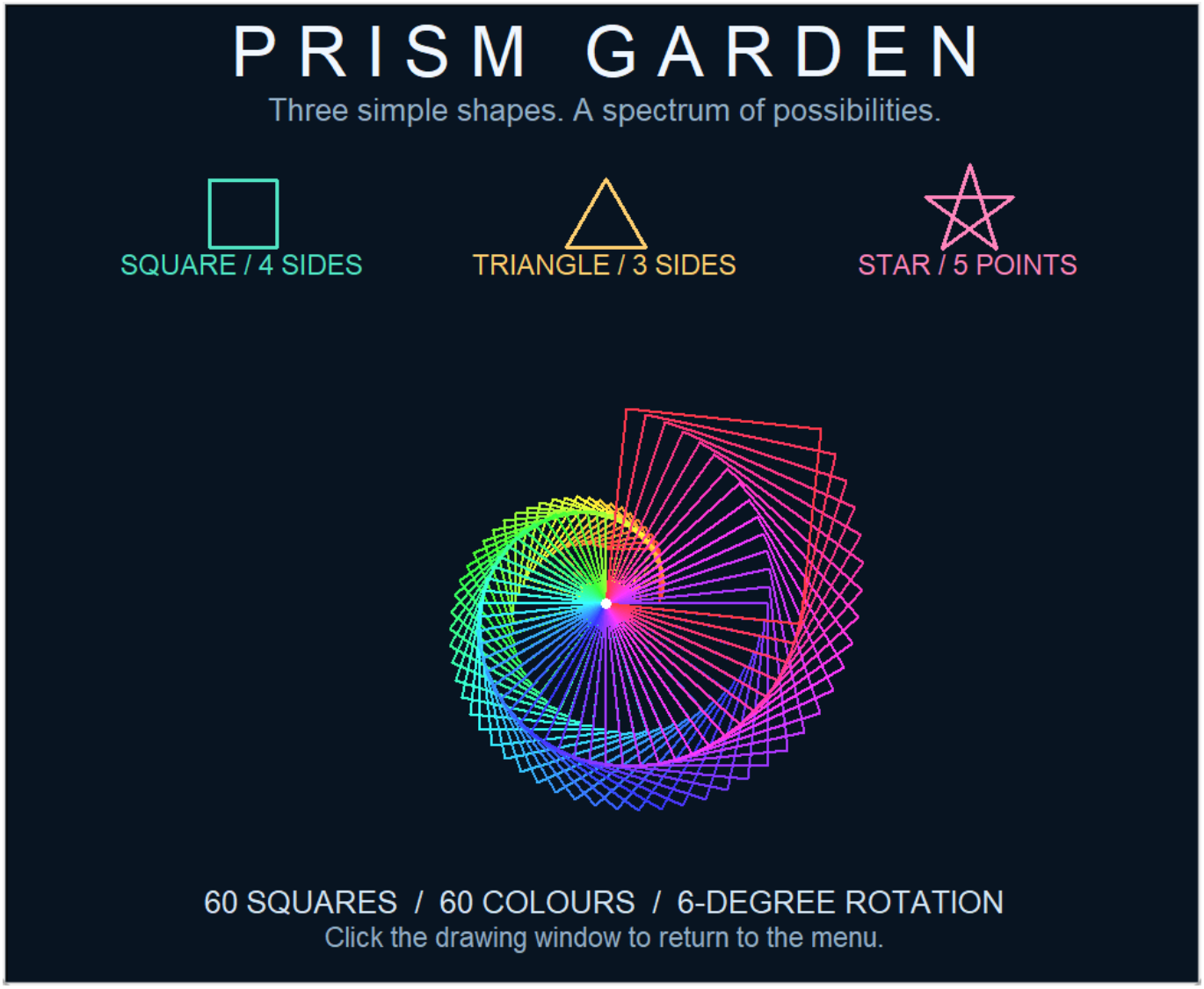


Figure 1. Captured output from the built-in Python turtle graphics engine.

Shapes and structured loops

Square, triangle and five-point star.

Dynamic colour and length

60 squares; HSV colours; lengths 45-163.

Run on Windows: `python "assignment 1_yourname.py"`

Run on Jetson: `python3 "assignment 1_yourname.py"`

Choose 1 for the drawing. Click its window to return to the menu.

02 Interactive QA Bot

Keyword matching with input() and if / elif / else



Figure 2. Screenshot of the actual qa_bot() execution transcript.
Predefined questions were supplied to input(); the responses were captured unchanged.

Requirement verification

Five required keywords	HKU, robot, jetson, ai, human
Question input	input() inside a repeating while loop
Conditional logic	Explicit if / elif / else branches
Input handling	Case-insensitive words; punctuation supported
Fallback and exit	Unknown topic, blank input, exit and quit
Validation	18 checks passed, including menu and calculator

Menu: 1 = drawing, 2 = QA Bot, 3 = optional calculator, 0 = exit.

When multiple topics occur, the first matching branch supplies the answer.

The drawing requires a graphical desktop with Tk support; no pip packages are needed.